



SIDDHARTH INSTITUTE OF ENGINEERING & TECHNOLOGY:: PUTTUR

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BIT BANK (OBJECTIVE)

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UNIT I

MATHEMATICAL LOGIC

1. A statement is a declarative sentence that is
 A) True B) False **C) True or False** D) None
 2. In the statement $P \rightarrow Q$, the statement P is called
 A) Consequent **B) Antecedent** C) Both A&B D) Sequent
 3. In the statement $P \rightarrow Q$, the statement P is called
A) Consequent B) Antecedent C) Both A&B D) Sequent
 4. A statement formula is an expression which is a string consisting of ____
 A) Variables B) Paranthesis C) Connective symbols **D) All of above**
 5. The proposition $(P \rightarrow Q) \wedge (Q \rightarrow P)$
 A) Tautology B) Contradiction **C) Contingency** D) Absurdity
 6. Which of the following statement is well formed formula
 A) $P \rightarrow Q \rightarrow \wedge Q$ B) $(P \wedge Q) \rightarrow R$ **C) $((Q \wedge (P \rightarrow Q)) \rightarrow R)$** D) None
 7. $((P \rightarrow Q) \vee \neg(P \rightarrow Q)) \wedge (P \rightarrow (P \rightarrow Q)) \Leftrightarrow$
 A) T B) F **C) Contingency** D) None
 8. A compound proposition that is neither a tautology nor a contradiction is called a _____
A) Contingency B) Equivalence C) Condition D) Inference
 9. $P \uparrow Q \Leftrightarrow$
 A) $P \wedge Q$ B) $\neg(P \vee Q)$ **C) $\neg(P \wedge Q)$** D) $P \vee Q$
 10. $P \downarrow Q \Leftrightarrow$
 A) $P \wedge Q$ **B) $\neg(P \vee Q)$** C) $\neg(P \wedge Q)$ D) $P \vee Q$
 11. If $p \rightarrow q$ is an implication ,then the implication $\neg q \rightarrow \neg p$ is called its
 A)Converse **B)Contrapositive** C)Inverse D)Alternative
 12. If $p \rightarrow q$ is an implication ,then the implication $p \rightarrow q$ is called its
A)Converse B)Contrapositive C)Inverse D)Alternative
 13. If $p \rightarrow q$ is an implication ,then the implication $\neg p \rightarrow \neg q$ is called its
 A)Converse B)Contrapositive **C)Inverse** D)Alternative
 14. $P \vee \neg P =$
 A) P **B)T** C)F D) ~P
 15. The connectives $\wedge$ and $\vee$ are also called ----- to each other
 A)NAND B) NOR C)XOR **D)dual**
 16. The proposition $p \wedge p$ is equivalent to
 A)1 **B) p** C) $\neg p$ D)None
 17. A proposition obtained by inserting the word not in the appropriate place is called
 A) Conjunction B) Disjunction **C)Negation** D)Implication
 18. If p is true , q is false then $p \rightarrow q$ is

- A) True **B) False** C) True or False D) None
 19. The logical truth or a universal valid statement is called []
 A) Contingency **B) Tautology** C) Absurdity D) Contradiction
 20. If $p(x)$: x is a prime number, then which of the following statements is true []
 A) $p(4)$ **B) $p(2)$** C) $p(9)$ D) $p(21)$
 21. Max-terms of two statements are formed by introducing the connective _____ []
 A) Conjunction **B) Disjunction** C) Negation D) Implication
 22. Min-terms of two statements are formed by introducing the connective _____ []
A) Conjunction B) Disjunction C) Negation D) Implication
 23. A formula consisting of a sum of elementary product is called []
 A) CNF **B) DNF** C) PDNF D) PCNF
 24. A Formula of disjunctions of Minterms only is known as []
 A) DNF B) CNF **C) PDNF** D) PCNF
 25. A formula consisting of a product of elementary sum is called []
A) CNF B) DNF C) PDNF D) PCNF
 26. A Formula of conjunctions of Maxterms only is known as []
 A) DNF B) CNF C) PDNF **D) PCNF**
 27. If a normal form contains all minterms, then it is a _____. []
 A) Contingency **B) Tautology** C) Contradiction D) Both A and B
 28. What rule of inference is used here? []
 "It is cloudy and drizzling now. Therefore, it is cloudy now."
 A) Addition B) Resolution C) Conjunction **D) Simplification**
 29. The Rule CP is also called []
 A) Contradiction of proof **B) Conditional proof** C) Consistency of premises D) None
 30. If $H_1, H_2, \dots, H_m$ are the premises and their conjunction is identically false then []
 the formulas $H_1, H_2, \dots, H_m$ are called
 A) Consistent B) Tautology **C) Inconsistent** D) None
 31. Let P : He is old q : He is clever, write the statement "He is old but not clever" in symbolic form []
 A) $p \wedge q$ **B) $p \wedge \neg q$** C) $\neg p \wedge \neg q$ D) $\neg(\neg p \wedge \neg q)$
 32. How many types of quantifiers are there in predicate logic? []
 A) 1 **B) 2** C) 3 D) 4
 33. In case of existential quantifier, the proposition $p(x)$ over the universe U is denoted by ____? []
 A) $x \exists p(x)$ B) $p(x) \exists x$ C) $p(x) x \exists$ **D) $\exists x p(x)$**
 34. In case of universal quantifier, the proposition $p(x)$ over the universe U is denoted by ____? []
 A) $x \forall p(x)$ B) $p(x) \forall x$ C) $p(x) x \forall$ **D) $\forall x p(x)$**
 35. A _____ quantifies a variable of a predicate? []
 A) Proposition B) Quantity **C) Quantifier** D) Quality
 36. Symbolize the statement "Every apple is red" []
 A) $(\exists x)(A(x) \wedge R(x))$ B) $(\forall x)(A(x) \wedge R(x))$
C) $(\exists x)(A(x) \rightarrow R(x))$ D) $(\forall x)(A(x) \rightarrow R(x))$
 37. $\neg(\forall x)A(x)$ []
 A) $(\forall x)A(x)$ B) $\neg(\exists x)A(x)$ C) $(\exists x)\neg A(x)$ D) None
 38. Multiple quantifiers can be used to quantify propositions with _____ variables? []
 A) Two **B) More than two** C) One D) none
 39. Which is the rule for the negation of quantified proposition? []
 A) Closure law B) Associative law **C) Demorgan's law** D) Identity law
 40. The symbolic form of "All men are mortal" where $M(x)$: x is a men $H(x)$: x is mortal []
 A) $Mx) \rightarrow H(x)$ **B) $(x)[M(x) \rightarrow H(x)]$** C) $(\exists x)(M(x) \rightarrow H(x))$ D) none

UNIT-II

SET THEORY

1. The numbers between 1 and 520, including both, are divisible by 2 or 6 is _____ []
A) 349 B) 54 C) 213 D) 303
2. The sum of all integers from 1 to 520 that are multiples of 4 or 5? []
A) 187 B) 208 C) 421 D) 52
3. In class, students want to join sports. 15 people will join football, 24 people will join basketball, and 9 people will join both. How many people are there in the class? []
A) 19 B) 82 C) 64 D) 30
4. In Inclusion-Exclusion Principle, if A and B are any two finite sets then -? []
A) $n(A \cap B) = n(A) + n(B) - n(A \cup B)$ B) $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
C) $n(A \cup B) = n(A) + n(B) - n(A \cap B)$ D) $n(A \cup B) = n(A) + n(B) + n(A \cap B)$
5. A pigeonhole is occupied by more than one pigeon if n pigeonholes are occupied by ____ or more pigeons? []
A) n B) n+1 C) n-1 D) 2n
6. The minimum number of students in a class such that three of them are born in the same month? []
A) 25 B) 82 C) 64 D) 30
7. A Poset in which every pair of elements has both least upper bound and greatest lower bound is termed as []
A) Sub lattices B) Lattices C) Trail D) Walk
8. _____ and _____ are the two binary operations defined for lattices. []
A) Join, meet B) Addition, subtraction
C) Union, intersection D) Multiplication, modulo division
9. Axioms that lattice L holds are – []
A) Commutative law B) Associative law C) Absorption law D) All of the above
10. Let $A = \{1, 2, 3, 4\}$. Let f, g and h be functions of A into R. Which one of them is one-one? []
A) $f(1) = 3, f(2) = 4, f(3) = 5, f(4) = 3$ **B) $g(1) = 2, g(2) = 4, g(3) = 5, g(4) = 3$**
C) $h(1) = 2, h(2) = 4, h(3) = 3, h(4) = 2$ D) $g(1) = 2, g(2) = 4, g(3) = 5, g(4) = 5$
11. Let $A = [-1, 1]$. Which of these functions are bijective on A? []
(A) $f(x) = x^2$ **(B) $g(x) = x^3$** (C) $h(x) = x^4$ (D) None
12. Let $S = \{a, b, c, d\}$. Which of the following sets of ordered pairs is a function of S into S? []
A) $\{(a, B), (c, A), (b, D), (d, C), (c, D)\}$ B) $\{(a, C), (b, C), (d, A), (c, B), (b, D)\}$
C) $\{(a, C), (b, D), (d, B)\}$ **D) $\{(d, B), (c, A), (b, D), a, C\}$**
13. If $x_1 = x_2 \Rightarrow f(x_1) = f(x_2)$ then the function f is said to be []
A) Injective B) Surjective C) Bijective D) Both A & B
14. If every element of y has the pre-image in x under the function of f then f is []
A) one-one **B) on-to** C) one-to-one D) none
15. If f^{-1} exists for 'f' then obviously f^{-1} is also []
A) one-one B) on-to **C) one-one & on-to** D) none
16. If $f(x) = x^2 + 1$ & $g(x) = x - 1$ then $f \circ g(x) =$ []
A) $x^2 - 2x + 2$ B) $x^2 - 2x - 2$ C) $x^2 - 2x$ D) $x + 1$
17. A mapping $I_x: X \rightarrow X$ is called an []
A) Reflexive B) identity C) inverse D) Inverse
18. Let g be a homomorphism from (X, o) to (Y, *). If $g: X \rightarrow Y$ is one-to-one and onto then g is called []
A) Bijection **B) Isomorphism** C) Epimorphism D) Monomorphism
19. Given $f(x) = x^3$ and $g(x) = x + 2$, for $x \in R$ then $f \circ g$ is []
A) $x + 2$ B) $x^3 + 2$ **C) $(x + 2)^3$** D) $x - 2$

20. Let $f: R \rightarrow R$ be given by $f(x) = x^3 - 2$. Find f^{-1} ? []
- A) $(x + 2)^{\frac{1}{3}}$ B) $(x - 2)^{\frac{1}{3}}$ C) $x^3 + 2$ D) $x^3 - 3$
21. The inverse of the identify element is the []
- A) Inverse element **B) Identity element** C) Idempotent element D) nilpotent element
22. If the set contains n elements, then the number of subsets is []
- A) n B) $n+1$ **C) 2^n** D) 2^{n+1}
23. If $A = \{1, 3, 4\}$, $B = \{1, 2, 3, 4, 5\}$ then []
- A) $A = B$ B) $B \subseteq A$ **C) $A \subseteq B$** D) None
24. The set of binary digits in tabular form is []
- A) $\{1, 1\}$ **B) $\{0, 1\}$** C) $\{0, 0\}$ D) $\{0, 1, 0\}$
25. If $B = \{x / x \text{ is a multiple of } 4, x \text{ is odd}\}$, the set B is []
- A) Proper subset B) Power set **C) Empty set** D) Index set
26. The family of subsets of any set is called as []
- A) Proper subset B) Subset C) Set of sets **D) Power set**
27. The algebraic system (S, o) is called _____ if the operation o is associative []
- A) Group B) Monoid **C) Semi group** D) Abelian group
28. If $(I, +)$ is a Monoid where I is the set of integers and $+$ is the operation of addition then the identity element is []
- A) 1 **B) 0** C) -1 D) 2
29. A group with addition binary operation is known as []
- A) Abelian group B) Groupoid C) Subgroup **D) Additive group**
30. A group with multiplication binary operation is known as []
- A) Abelian group B) additive group **C) multiplicative group** D) Closure
31. A group G is said to be _____ if the commutative law holds []
- A) Groupoid B) Semi group **C) Abelian** D) Closure
32. In order word (S, o) is a semi group if for any $x, y, z \in S$ then $x o (y o z) = (x o y) o z$ []
- A) $(x o y) * z$ B) $(x o z) o y$ **C) $(x o y) o z$** D) $x * (y * z)$
33. Semi group homomorphism satisfies []
- A) on-to B) one-one C) one-one & on-to **D) none**
34. Every homomorphic image of an abelian group is []
- A) Sub group B) Semigroup **C) Abelian group** D) group
35. A non-empty subset H of a group $(G, *)$ a subgroup iff _____ where $a \in H, b \in H$ []
- A) $ab \in H$ B) $a * b \in H$ **C) $a * b^{-1} \in H$** D) $a^{-1} * b \in H$
36. The identity element (if it exists) of any algebraic structure is []
- A) multiple **B) unique** C) one D) zero
37. The non -zero set of intergers under multiplication is []
- A) monoid** B) semigroup C) group D) Abelian group
38. The inverse element of 4 in the group $(Z, +)$ []
- A) 3 B) 5 **C) -4** D) +4
39. The group of all one- one & onto mappings from S to S there the order of S is n , and is called agroup. []
- A) an abelian B) symmetric **C) alternating** D) commutative
40. The inverse element of 5 in the group $(Z, +)$ []
- A) 3 **B) -5** C) -4 D) +4

UNIT- III

ELEMENTARY COMBINATORICS

1. Enumerating r-permutations without repetitions $P(n,r)=$ [c]
A) $\frac{n!}{r!(n-r)!}$ B) $\frac{n!}{r!}$ C) $\frac{n!}{(n-r)!}$ D) $\frac{r!}{(n-r)!}$
2. How many 3 digit number can be formed using the digits 1,3,4,5,6,8 and 9 []
A) $7*6*5$ B) $3!$ C) $\frac{7!}{3!}$ D) 7^3
3. How many 5-card hands have 2clubs and 3hearts. []
A) $C(13,2) C(12,3)$ B) **$C(13,2) C(13,3)$** C) $C(52,5)$ D) None
4. If a student is to answer true or false questions and there are five questions, the number of ways, he can answer is []
A) **10** B) 16 C) 32 D) 5
5. The number of one-digit numbers can be formed from the digits 1,2,3,4,5,6 is []
A) 5 B) **6** C) 2 D) 0
6. The four-digit numbers, that can be formed from the digits 1,2,3,4,5,6,7 if there will be no repetitions are allowed is []
A) 24 B) 6 C) **840** D) 120
7. The three-digit numbers, that can be formed from the digits 1,2,3,4,5 if repetitions are allowed is []
A) **125** B) 120 C) 60 D) 36
8. The number of ways of selecting an ace card is []
A) 2 B) 3 C) **4** D) 5
9. The number of ways of drawing 2 cards with replacement from a deck of 52 cards is []
A) **2704** B) 1326 C) 52 D) 2652
10. The number of ways of drawing 2 cards without replacement from a deck of 52 cards is []
A) 2704 B) 1326 C) 52 D) **2652**
11. There are 12 red balls and 8 blue balls in a box. The number of ways of selecting 5 red balls and 3 blue balls is []
A) 42126 B) **44352** C) 12118 D) 24352
12. The number of positive integer solutions of $x+y+z=6$ is []
A) 24 B) 20 C) **10** D) 15
13. The number of two digit even number is []
A) 45 B) 24 C) 81 D) **50**
14. The three-digit numbers, that can be formed from digits 1,2,3,4,5, if repetitions are not allowed is []
A) 125 B) **60** C) 45 D) 90
15. The number of non-negative integer solutions of $x+y+z=6$ is []
A) 24 B) 20 C) 60 D) **28**
16. The number of non-negative integer solutions of $x+y+z=9$ is []
A) **55** B) 45 C) 60 D) 72
17. The number of positive integer solutions of $x+y+z<7$ is []
A) 20 B) 60 C) **120** D) 90
18. The number of permutations of the word SUCCESS is []
A) 960 B) **420** C) 120 D) 840
19. The number of permutations of the word HAPPY is []
A) 90 B) 120 C) **60** D) 40
20. The number of permutations of the word LAPTOP is []
A) 240 B) 120 C) **360** D) 40030
21. The number of combinations of five objects among eight objects, if the repetitions are allowed and

- order is not important is []
 A)645 **B) 792** C) 896 D) 962
22. The number of combinations of three objects among six objects, if the repetitions are allowed and order is not important is []
 A)**56** B)96 C) 48 D) 120
23. There are two groups, each consists of four questions each. If a student is to answer 2 from one group and 3 from another group, the number of ways that he can answer is []
 A)48 **B)24** C)72 D) 30
23. The coefficient of x^5y^2 in the expansion of $(x+2y)^7$ is []
 A)42 B) 84 C) 120 D) 96
24. The coefficient of x^5y in the expansion of $(2x+y)^6$ is []
 A)192 **B) 128** C) 120 D) 144
25. $P(5,3)=$ _____ []
 A) 60 B)61 C)70 D)50
26. $P(n,3) = P(n,2)$ then $n=$ _____ []
 A) 2 B)3 C)1 D)4
27. $C(n,5) = C(n,6)$ then $n=$ _____ []
 A)11 B)10 C)12 D)13
28. The number of non negative integer solutions of $x_1 + x_2 + x_3 + x_4 = 8$ have..... []
 A)165 B)164 C) 166 D)163
29. The coefficient of x^4y^7 in the expansion of $(x - y)^{11}$ is []
 A)-330 B) 330 C) - 332 D) 332
30. The number of non negative integer solutions of $x_1 + x_2 + x_3 = 11$ have..... []
 A)65 B)74 C) 75 D)78
31. The coefficient of x^2y^2 in the expansion of $(2x + 3y)^{10}$ is []
 A)1620 B)162 C)1820 D)1520
32. The coefficient of x^3y^2 in $(x + y)^5$ []
 A) $C(5,3)$ B) $C(5,2)$ C) $C(5,5)$ D) $C(2,5)$
33. The coefficient of $x^{n_1}y^{n_2}z^{n_3}$ in $(x + y + z)^n$ []
 A) $\frac{n!}{n_1!n_2!n_3!}$ B) $\frac{n!}{x_1!y_2!z_3!}$ C) $\frac{n!}{3!}$ D) $\frac{n!}{(xyz)!}$
34. By solving $C(n, 2) = 28$, $n =$ []
 A) 9 B) 8 C) 7 D) 10
35. The number of circular permutations of n objects taken all n at a time is []
 A) $n - 1$ B) $(n - 1)!$ C) n D) $n!$
36. If anti clock wise & clock wise order of arrangements are not distinct then the number of circular permutations of a distinct items is []
 A) $N - 1$ B) $(n - 1)!$ C) $\frac{1}{2} (n - 1)!$ D) none
37. The coefficient of $x^2y^3z^2$ in the expansion of $(x + y + z)^7$ is []
 A)120 B)200 C) 820 D)210
38. By solving $C(n, 2) = 36$, $n =$ []
A) 9 B) 8 C) 7 D) 10
39. If $C(n,1) = C(n, 2)$ then $n =$ []
 A)2 B) 1 C) 3 D) 4
40. The coefficient of $x^2y^2z^2$ in the expansion of $(x + y + z)^6$ is []
 A)90 B)100 C) 80 D)10

UNIT IV

RECURRENCE RELATIONS

1. The series $1+x+x^2+\dots =$ []
 A) $\sum x^r$ B) $\sum (-1)^r x^r$ C) $\sum (-A)^r x^r$ D) None
2. The co-efficient of x^{27} in the expansion of $(x^4+x^5+x^6+\dots)^5$ is----- []
 A) 300 B) 280 C) 200 **D) 330**
3. The solution of linear recurrence relation has ----methods []
 A) 4 **B) 3** C) 2 D) None
4. Iteration method is also called as-----method []
 A) Not substitution B) Characteristic root **C) Step by step** D) None
5. In which method ,the solution is obtained as the sum of two parts---- []
 A) Substitution **B) Characteristic root** C) Step by step D) None
6. When $f_n = 0$, then the equation $c_0 a_n + c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k} = f(n)$ is----- []
A) Homogeneous B) Non-Homogeneous C) Step by step D) None
7. If the characteristic equation has 2, -1 roots, then the solution is---- []
A) $a_n = b_1 2^n + b_2 (-1)^n$ B) $a_n = (b_1 + 2b_2)(-1)^n$ C) $(2b_1 + (-1)b_2)r^n$ D) None
8. The solution of linear non-homogeneous equation is----- []
A) $a_n = a_n^{(h)} + a_n^{(p)}$ B) $a_n = A_0 + A_1 n + A_2 n^2$ C) $A b^n$ D) None
9. -----is called a particular solution []
 A) $a_n^{(h)}$ **B) $a_n^{(p)}$** C) $a_n = a_n^{(h)} + a_n^{(p)}$ D) None
10. $a_n = 2a_{n-1}$ is a homogeneous linear recurrence relation of order----- []
 A) 2 B) 3 **C) 1** D) None
11. If $f(n) = 2^n$ and 2 is the root of the characteristic equation, then the trial solution is --- []
A) $A 2^n n^2$ B) $A 2^{2n^2}$ C) $A 2^2 n^2$ D) None
12. The solution of linear homogeneous recurrence relation is---- []
A) $a_n^{(h)}$ B) $a_n^{(p)}$ C) a_n D) None
13. $\sum a_n x^n$ is equal to----- []
A) $a_0 + a_1 x + a_2 x^2 + \dots$ B) $a_0 x + a_1 x^2 + a_2 x^3 + \dots$ C) $a_0 + a_1 x$ D) None
14. A recurrence relation is a formula that relates for any integer----- []
A) $n \geq 1$ B) $n \leq 1$ C) $n = 0$ D) None
15. If the solution is $a_n = (b_1 + b_2 n + b_3 n^2) 2^n$, then the value of “r” is----- []
 A) 3 **B) 2** C) 1 D) None
16. If $f(n)$ is constant then the trial solution is --- []
 A) $A b^n$ **B) A** C) $A b^n s^n$ D) None
17. Solving recurrence relation for ----types []
 A) 2 **B) 3** C) 1 D) None
18. If $a_k = 2a_{k-1} + k$, for all $k \geq 2, a_1 = 1$, then the value of $a_3 =$ ----- []
 A) 12 **B) 11** C) 4 D) None
19. If $a_{n+2} - 4a_{n+1} + 4a_n = 2^n$, then the equation is []
 A) Homogeneous **B) Non-Homogeneous** C) Characteristic D) None
20. Trail solution of $a_n^{(p)}$ is $A_0 + A_1 n + A_2 n^2 + \dots + A_m n^m$, then the degree is----- []
 A) 2 **B) m** C) n D) None
21. The generating function of 1 is []
A) $\frac{1}{1-x}$ B) $\frac{1}{1+x}$ C) $\frac{1}{1-2x}$ D) $\frac{x}{1-2x}$
22. The generating function for the sequence 1,2,3,4,... []
 A) $\frac{x}{1-3x}$ B) $\frac{x}{1+3x}$ C) $\frac{1}{1+x}$ **D) $\frac{1}{(1-x)^2}$**
23. The generating function of n is []
 A) $\frac{1}{1+x}$ B) $\frac{1}{1-x}$ **C) $\frac{x}{(1-x)^2}$** D) $\frac{1}{(1-x)^2}$
24. The generating function of $1+n$ is []
 A) $\frac{1}{1-x}$ B) $\frac{1}{1+x}$ C) $\frac{x}{(1-x)^2}$ **D) $\frac{1}{(1-x)^2}$**

25. The generating function of the sequence 1, -2, 4, -8, 16is []
 A) $\frac{x}{1+2x}$ B) $\frac{1}{1+2x}$ C) $\frac{x}{(1-x)^2}$ D) $\frac{x^2}{(1+2x)^2}$
26. The exponential generating function of the sequence 1,1,1,1.....is []
 A) e^x B) e^{-x} C) e^{2x} D) e^{-2x}
27. The exponential generating function of the sequence 1,0,-1,0,1,0,-1,0,1,.....is []
 A) $\cos x$ B) $\sin x$ C) $\cos 2x$ D) e^{2x}
28. $1+x+x^2+x^3+\dots =$ []
 A) $\frac{1}{1+x}$ B) $\frac{1}{1+x^2}$ C) $\frac{1}{(1-x)^2}$ D) $\frac{1}{(1-x)}$
29. The order of RR $a_{n+1} - 2a_n = 2$ is []
 A) 2 B) 1 C) 3 D) 4
30. The order of $a_{n-2} + a_{n-1} + a_n$ is..... []
 A) 1 B) 2 C) 3 D) 4
31. The order of $a_{n+2} - a_{n+1} = 2a_n$ is []
 A) 3 B) 2 C) 1 D) 4
32. The characteristic equation has 1,1 roots, then the solution is []
 A) $a_n = b_1 + nb_2$ B) $a_n = b_1 + b_2$ C) $a_n = b_1 - nb_2$ D) $a_n = b_1 - b_2$
33. What is the recurrence relation for the sequence 5,8,11,14.....? with $a_0=5$ []
 A) $a_n = 3a_{n-1} - 2a_{n-2}$ B) $a_n = 3a_{n-1} - 2a_{n-2}$ C) $a_n = a_{n-1} + 3$ D) $a_n = 3a_{n-1} - 2a_{n-3}$
34. Find the value of a_4 for the recurrence relation $a_n = 2a_{n-1} + 3$ with $a_0=6$ []
 A) 141 B) 221 C) 320 D) 65
35. The order of $a_n = -3a_{n-1}$ is..... []
 A) 3 B) 2 C) 0 D) 1
36. What will be the sequence generated by the generating function $\frac{4x}{(1-x)^2}$ []
 A) 12,16,20,24,... B) 1,3,5,7,9,... C) 0,4,8,12,16,20,.. D) 0,1,1,3,5,8,13...
37. Given that $a_0=1$, $a_n = n + (-1)^n a_{n-1}$ for $n \geq 2$. What is the value of a_4 ? []
 A) 1 B) 4 C) 5 D) 8
38. The characteristic equation has 2,4,4 roots then the solution is []
 A) $a_n = b_1 2^n + (b_2 + nb_3) 4^n$ B) $a_n = b_1 4^n + b_2 4^n + nb_3 4^n$
 C) $a_n = b_1 2^n + (nb_2 + n^2 b_3) 4^n$ D) $a_n = b_1 2^n + (b_2 + b_3) 4^n$
39. For the recurrence relation $a_{n+2} - 5a_{n+1} + 6a_n = 2$, the value of $a_n^{(h)}$ is []
 A) $3b_1 + 2b_2$ B) $b_1 3^n + b_2 2^n$ C) $b_1 + 5b_2$ D) $b_1 + b_2$
40. If $f(n) = (-3)^n$ and (-3) is not root of the characteristic equation then the trial solution is []
 A) $A(-3)^n$ B) $(-3)^n$ C) $A^2(-3)^n$ D) $A^n(-3)^n$

UNIT-V **GRAPHS**

1. A regular graph of degree _____ has no lines. []
A) 0 B) 1 C) 2 D) 3
2. The maximum degree of any vertex in a simple graph with n vertices is []
A) n B) n+1 **C) n-1** D) n+2
3. A graph G has 21 edges, 3 vertices of degree 4 and other vertices of degree 3. Find the number of vertices in G. []
A) 10 B) 11 C) 12 **D) 13**
4. The maximum number of edges in a simple graph with n vertices is []
A) $n(n-1)/2$ B) $(n-1)/2$ C) $n(n+1)/2$ D) $n(n1)$
5. A graph which allows more than one edge to join a pair of vertices is called []
A) Simple graph **B) Multi-graph** C) Null graph D) Weighted graph
6. A graph G with no self loops is called a []
A) Simple graph **B) Multi-graph** C) Null graph D) Weighted graph
7. A graph having loops and multiple edges called []
A) Simple graph B) Multi-graph **C) Pseudo graph** D) Weighted graph
8. A simple graph G, in which every pair of distinct vertices are adjacent is called []
A) Simple graph B) Multi-graph C) Null graph **D) Complete graph**
9. A binary tree T has n leaves. The number of nodes of degree 2 in T is []
A) n-1 B) n C) n+1 D) 2n
10. The total number of edges of a complete graph K_n is... []
A)n B) n^2 C) $\frac{n(n+1)}{2}$ **D) $\frac{n(n-1)}{2}$**
11. A graph without edges is called agraph []
A) trivial graph **B) null graph** C) infinite graph D) simple graph
12. A graph is regular , if the degree of each vertex is []
A) same B) not same C) always zero D) always one
13. A graph with order 0 or 1 is _____ []
A) Null graph **B) Trivial graph** C) Simple Graph D) Finite graph
14. A regular graph of degree _____ has no lines. []
A) 0 B) 1 C) 2 D) 3
15. A graph with both directed and undirected edges is _____ []
A) Finite graph B) Directed graph **C) Mixed graph** D) Undirected graph
16. A graph G has 21 edges, 3 vertices of degree 4 and other vertices of degree 3. Find the number of vertices in G. []
A) 10 B) 11 C) 12 **D) 13**
17. The maximum degree of any vertex in a simple graph with n vertices is []
A) n B) n+1 **C) n-1** D) n+2
18. Euler's rule is []
A) $v+e+r=2$ **B) $v-e+r=2$** C) $ve-r=2$ D) $v+er=2$
19. A planar graph has only ____ infinite region(s). []
A) one B) two C) Three D) four
20. If a connected planar graph G has e edges, v vertices and r regions, then []
A) $v+e+r=2$ **B) $v-e+r=2$** C) $ver=2$ D) $v+er=2$
21. A connected graph that contains an Euler Circuit is called []
A) Euler trail B) Semi-Euler graph **C) Euler graph** D) Hamilton graph
22. A graph has vertices of degree 0,2,2,3,9 How many edges does the graph have []
A) 8 B) 9 C) 10 D) 11
23. A graph $G=(V,E)$ is called a ____ graph if its vertices V can be partitioned into two subsets V_1 and V_2 such that each edge of G connects a vertex of V_1 to a vertex of V_2 []
A) simple B) bipartite **C) complete bipartite** D) multi graph

24. The no of vertices of a graph is _____ []
 A) size **B) order** C) loop D)pendent
25. A complete graph with n vertices will have _____ edges []
 A) $(n-1)(n-2)/2$ **B) $n(n-1)/2$** C) $(n-2)/2$ D) $n(n-2)/2$
26. A graph which allows more than one edge to join a pair of vertices is called a []
 A) simple graph B) null graph **C) multi graph** D) Pseudo graph .
27. If G is a connected graph with n vertices and m edges, a spanning tree of g must have__ edges []
 A) n B) n+1 C) n+3 **D) n-1**
28. A given connected graph is a Euler graph if and only if all vertices of G are of []
 A) same degree **B) even degree** C) odd degree D) Different degree
29. An _____ through a graph is a path whose edge list contains each edge of the graph exactly once. []
 A) Euler path **B) Euler circuit** C) Euler graph D) Euler region
30. An _____ is a graph that possesses a Euler circuit. []
 A) Euler path B) Euler circuit **C) Euler graph** D) Euler region
31. A circuit in a connected graph which includes every vertex of the graph is known as []
A) Euler B) Universal C) Hamiltonian D) Clique
32. If G is agraph within vertices, then a Hamiltonian cycle in G will contain exactly _____ edges. []
 A) n-1 B) n **C) n+1** D) n+2
33. A complete graph of five vertices is _____ []
 A) Planar **B) Non planar** C) Sub graph D) Euler graph
34. A circuit in a connected graph which includes every vertex of the graph is known as []
 A) Euler B) Universal **C) Hamiltonian** D) Clique
35. The number of edges of a graph is _____ []
 A) branch B) loop C) order **D) size**
36. If the degree of the vertex is one then the vertex is _____ []
 A) Infinite vertex B) finite vertex C) pseaduo vertex **D) Pendent vertex**
37. A graph which contains only isolated node is _____ []
 A) Regular graph **B) Null graph** C) Complete graph D) None
- 38.A tree with only one vertex is called _____ []
A) Trivial tree B) Rooted tree C) Branch D) Node
39. A rooted tree in which each internal vertex has atmost two children is called _____ []
 A) Spanning tree **B) Binary tree** C) Minimal tree D) Maximal tree
40. A tree with n vertices have _____ edges. []
 A) $n(n+1)$ B) n+1 **C) n-1** D) $n(n-1)$